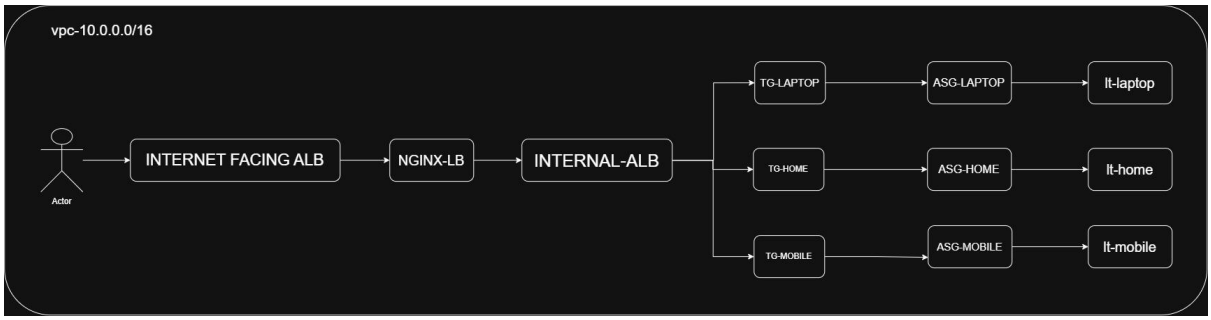
INTERNAL AND INTERNET FACING LOAD BALANCER

ARCHITECTURE



PHASE-1 (creation of vpc)

1. create a vpc (10.0.0.0/16)
2. create a subnets
    1. public-subnet-1 – (10.0.1.0/24) – (az-1a)
    2. public-subnet-2 – (10.0.2.0/24) – (az-1b)
    3. private-subnet-1 – (10.0.11.0/24) – (az-1a)
    4. private-subnet-2 – (10.0.12.0/24) – (az-1b)
3. create a internet gateway and attach to a vpc
4. create a nat gateway in public-subnet-1
5. create a route table
    1. public-rt
        1. associate all public subnets
        2. add internet gateway to routes
    2. private-rt
        1. associate all private subnets
        2. add nat gateway to routes
6. Edit DNS hostname of vpc
    1. click on vpc
    2. click on actions
    3. edit vpc settings
    4. enable dns hostnames
    5. save

PHASE-2 (creation of security groups)

1. SG-INTERNET-ALB

1. Allow http -> 0.0.0.0/0
2. Allow https -> 0.0.0.0/0

2. SG-NGINX
    1. Allow http -> SG-INTERNET-ALB

3. SG-INTERNAL-ALB
    1. Allow 8001 -> SG-NGINX
    2. Allow 8002 -> SG-NGINX
    3. Allow 8003 -> SG-NGINX

4. SG-APPLICATION
    1. Allow 8001 -> SG-INTERNAL-ALB
    2. Allow 8002 -> SG-INTERNAL-ALB
    3. Allow 8003 -> SG-INTERNAL-ALB

PHASE-3 (creation of launch template)

1. LT-HOME
    1. name – (LT-HOME)
    2. ami – (amazon linux)
    3. instance_type – t3.micro
    4. sg - SG-APPLICATION
    5. user_data

```
#!/bin/bash
dnf update -y
dnf install httpd -y
sed -i 's/^Listen 80$/Listen 8001/' /etc/httpd/conf/httpd.conf
systemctl enable httpd
systemctl start httpd
echo "<h1>Home instance $HOSTNAME</h1>" > /var/www/html/index.html
systemctl restart httpd
```

2. LT-MOBILE
    1. name – (LT-MOBILE)
    2. ami – (amazon linux)
    3. instance_type – t3.micro
    4. sg - SG-APPLICATION
    5. user_data

```
#!/bin/bash
dnf update -y
dnf install httpd -y
sed -i 's/^Listen 80$/Listen 8002/' /etc/httpd/conf/httpd.conf
```

```
systemctl enable httpd
systemctl start httpd
mkdir -p /var/www/html/mobile
echo "<h1>Mobile instance $HOSTNAME</h1>" > /var/www/html/mobile/index.html
systemctl restart httpd
```

3. LT-LAPTOP
    1. name – (LT-LAPTOP)
    2. ami – (amazon linux)
    3. instance_type – t3.micro
    4. sg - SG-APPLICATION
    5. user_data

```
#!/bin/bash
dnf update -y
dnf install httpd -y
sed -i 's/^Listen 80$/Listen 8003/' /etc/httpd/conf/httpd.conf
systemctl enable httpd
systemctl start httpd
mkdir -p /var/www/html/laptop
echo "<h1>Laptop instance $HOSTNAME</h1>" > /var/www/html/laptop/index.html
systemctl restart httpd
```

PHASE-4 (creation of autoscaling group)

ASG-HOME

1. Name – ASG-HOME
2. Launch template – LT-HOME
3. Vpc – my-vpc
4. Subnets – private-subnet-1/private-subnet-2
5. Desired capacity – 1
6. Minimum capacity – 1
7. Maximum capacity – 4
8. Enable health check - EC2 health check

ASG-MOBILE

1. Name – ASG-MOBILE
2. Launch template – LT-MOBILE
3. Vpc – my-vpc
4. Subnets – private-subnet-1/private-subnet-2
5. Desired capacity – 1
6. Minimum capacity – 1
7. Maximum capacity – 4
8. Enable health check - EC2 health check

ASG-LAPTOP

1. Name – ASG-LAPTOP
2. Launch template – LT-LAPTOP
3. Vpc – my-vpc
4. Subnets – private-subnet-1/private-subnet-2
5. Desired capacity – 1
6. Minimum capacity – 1
7. Maximum capacity – 4
8. Enable health check - EC2 health check

PHASE-5 (creation of target groups)

TG-HOME

1. Target type – Instance
2. Tg name – TG-HOME
3. Vpc – my-vpc
4. Port – 8001
5. Protocol – http
6. Health check path - /
7. Create a target group

TG-MOBILE

1. Target type – Instance
2. Tg name – TG-MOBILE
3. Vpc – my-vpc
4. Port – 8002
5. Protocol – http
6. Health check path - /mobile/
7. Create a target group

TG-LAPTOP

1. Target type – Instance
2. Tg name – TG-LAPTOP
3. Vpc – my-vpc
4. Port – 8003
5. Protocol – http
6. Health check path - /laptop/
7. Create a target group

PHASE-6 (internal application load balancer)

1. name – INTERNAL -ALB
2. lb type – application load balancer
3. scheme – internal
4. vpc – my-vpc
5. subnets – private-subnet-1/private-subnet-2
6. sg - SG-INTERNAL-ALB
7. listener – port (8001) – forward to tg (TG-HOME)
8. listener – port (8002) – forward to tg (TG-MOBILE)
9. listener – port (8003) – forward to tg (TG-LAPTOP)

PHASE-7 (creation of nginx-launch-template)

LT-NGINX

1. name – (LT-NGINX)
2. ami – (amazon linux)
3. instance_type – t3.micro
4. sg - SG-NGINX
5. user_data

```
#!/bin/bash
dnf update -y
dnf install nginx -y
cat > /etc/nginx/conf.d/app.conf <<'EOF'
server {
  listen 80;
  location / {
    proxy_pass http://internal-INTERNAL-ALB-1626036581.ap-south-
1.elb.amazonaws.com:8001;
    proxy_set_header Host $host;
    proxy_set_header X-Real-IP $remote_addr;
    proxy_set_header X-Forwarded-For $proxy_add_x_forwarded_for;
  }
  location /mobile/ {
    proxy_pass http://internal-INTERNAL-ALB-1626036581.ap-south-
1.elb.amazonaws.com:8002;
    proxy_set_header Host $host;
    proxy_set_header X-Real-IP $remote_addr;
    proxy_set_header X-Forwarded-For $proxy_add_x_forwarded_for;
  }

  location /laptop/ {
    proxy_pass http://internal-INTERNAL-ALB-1626036581.ap-south-
1.elb.amazonaws.com:8003;
    proxy_set_header Host $host;
    proxy_set_header X-Real-IP $remote_addr;
    proxy_set_header X-Forwarded-For $proxy_add_x_forwarded_for;
```

```
    }

}
EOF
nginx -t
systemctl enable nginx
systemctl start nginx
```

PHASE-8 (creation of nginx-autoscaling-group)

ASG-NGINX

1. Name – ASG-NGINX
2. Launch template – LT-NGINX
3. Vpc – my-vpc
4. Subnets – private-subnet-1/private-subnet-2
5. Desired capacity – 1
6. Minimum capacity – 1
7. Maximum capacity – 4
8. Enable health check - EC2 health check

PHASE-9 (creation of nginx-target-group)

TG-NGINX

1. Target type – Instance
2. Tg name – TG-NGINX
3. Vpc – my-vpc
4. Port – 80
5. Protocol – http
6. Health check path - /
7. Create a target group

Attach these target group to autoscaling group
ASG-NGINX → Integrations → Load balancing → TG → TG-NGINX

PHASE-10 (creation of internet-facing-LB)

1. name – INTERNET-FACING-ALB
2. lb type – application load balancer
3. scheme – internet-facing
4. vpc – my-vpc
5. subnets – public-subnet-1/public-subnet-2
6. sg - SG-INTERNET-ALB
7. listener – port (80) – forward to tg (TG-NGINX)
8. create a target group